Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption
One potential explanation for the website's connection timeout error message is: The web server is becoming overwhelmed by an abnormally large number of connection requests and is unable to respond to legitimate users. The logs show that: A single source with the IP address `203.0.113.0` begins sending SYN requests at log entry 52. The SYN requests continue in rapid succession, and the server gradually becomes unable to respond to legitimate employee traffic. This event could be: A **Denial-of-Service (DoS) SYN flood attack**. Because the malicious traffic originates from a single IP address, this is a DoS attack rather than a DDoS attack.

Section 2: Explain how the attack is causing the website to malfunction
When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake: **1. SYN:** The client sends a SYN packet requesting a connection with the web server. **2. SYN/ACK:** The server responds with a SYN/ACK and reserves resources for the connection. **3. ACK:** The client sends an ACK to complete the connection. Explain what happens when a malicious actor sends a large number of SYN packets all at once: Once the attacker begins sending large numbers of SYN packets in rapid succession, the server has to use resources handling those connection requests. As the requests continue, the server becomes overwhelmed and eventually has difficulty responding to legitimate users. Explain what the logs indicate and how that affects the server: The logs show repeated SYN requests from `203.0.113.0`. As the attack continues, legitimate employee connections begin receiving RST/ACK responses and 504 Gateway Time-out errors. Eventually, the web server becomes unresponsive to legitimate employee traffic.